

and can be derived from the squared Earth mover’s distance (EMD) by Hou et al. (?) for the ordinal basic human ratings prediction.

We show the predicted score histograms by our proposed CJS-CNN and other compared loss functions on the test set of AVA in the supplementary material. Our CJS-CNN achieves the most similar results to the ground truth human rating distributions.

Numerical Evaluation Results In Table 1, we summarize the evaluation results of the loss functions over the divergences. We use the Mean Divergences (MD) to evaluate various divergences between the predicted score histogram and the ground truth on the test set of AVA. The MD is defined as:

$$MD = \frac{1}{n} \sum_{i=1}^n l(y, \hat{y}), \quad (14)$$

where $l = \{l^{PED}, l^{PCE}, l^{PJS}, l^{PCS}, l^{PKL}, l^{CED}, l^{CJS}\}$ defined above. n is size of the test set.

The results in Table 1 reveal that, our proposed RS-CJS and CJS based CNN outperform other methods. Among all the odd lines with white background, our CJS achieves the best performance. All the mean divergences of our RS-CJS on the test sets are the smallest. Typically, in a learning setting, optimizing directly a certain criterion should lead to higher performance than optimizing a related one. However, although our methods are optimizing the CJS loss, the learned model can achieve best performance in other related loss. This is mainly because that, as verified by (?), the CJS is quite suitable for non-parametric computation on empirical data, such as the ordinal basic rating data of the AVA dataset. The line with header ‘RS-’ means adding our reliability-sensitive learning strategy. Almost all the RS version methods (the even lines) perform better than the corresponding ones (the odd lines). The reliability sensitive learning based on the kurtosis reduces the impacts of the unreliable training samples.

The Ablation Study of the Reliability Factor Wu et al. (?) propose to use the number of ratings of each image for the reliability factor $r^{ratnum}(y)$. The larger the rating number is, the larger the reliability of rating is. To compare with our kurtosis based reliability factor $r^{kurtosis}(y)$ in Eq. 6 and Eq. 7, we use an balance factor λ as follow to make ablation study.

$$r(y) = \lambda r^{kurtosis}(y) + (1 - \lambda) r^{ratnum}(y) \quad (15)$$

For a fair comparison, we use $r(y)$ on the CJS loss: $r(y)CJS(y, \hat{y})$

The comparison results are shown in Table 2. The results reveal that the performance of $r^{kurtosis}(y)$ are slightly better than that of $r^{ratnum}(y)$. The combination of these two reliability factors does not produce better performance. Note that, the kurtosis can be directly computed using the normalized score histogram. While the rating number is additional information of the normalized score histogram and is not always available in the training set.

Conclusions and Discussions

In this paper, we propose the CJS-CNN to predict the aesthetic score distribution of images. Unlike the object recognition, which definitely has right answers in most cases, the image aesthetic assessment is a subjective task in nature. Thus, only using a scalar to represent the aesthetics may not be the right direction.

Instead of only predicting the binary high-low label or the numerical score, we can output the aesthetic score distribution with rich statistics for various applications such as aesthetic image retrieval, aesthetic image enhancement. The overall aesthetic quality can be represented by the mean or median. The controversy or subjectiveness can be measured by the variance. The popularity of an image can be measured by the rating number of human. However, the rating number cannot be derived from the predicted score histogram. As shown in the experiments, we can use the kurtosis to approximate the popularity instead of the rating number.

The aesthetic quality assessment is a subjective task in nature. It has been a long time that people focused on the scalar representation (1D numerical or binary coding) of aesthetics. Wu et al. (?) pointed out this problem and made an attempt to predict the score distribution. However, it was submerged in rich literatures which aim to rise the classification or regression accuracy of the scalar representation. With the powerful deep representation learning technologies, we think it is the right time to let the aesthetic quality assessment return to it’s subjective nature. This paper is a restart of this direction. We hope it can inspire more work in the future, such as (1) mapping more statistics to the subjective evaluation of vast amount of images, (2) designing new large scale aesthetic datasets with unbiased data and specially for subjective assessment of aesthetics, (3) using more powerful and larger DCNNs or other machine learning technologies to make the assessment by computer better match that of human.

Acknowledgments

We thank all the reviewers and ACs. This work is partially supported by the National Natural Science Foundation of China (Grant Nos. 61402021, 61401228, 61402463, 61772513), the Science and Technology Project of the State Archives Administrator (Grant No. 2015-B-10), the open funding project of State Key Laboratory of Virtual Reality Technology and Systems, Beihang University (Grant No. BUAA-VR-16KF-09), the Fundamental Research Funds for the Central Universities (Grant No. 3122014C017), the China Postdoctoral Science Foundation (Grant No. 2015M581841), and the Postdoctoral Science Foundation of Jiangsu Province (Grant No. 1501019A).

A quisquam magni ea vel voluptatum quidem at dignissimos, dolor laudantium amet vel ad consectetur. Error repellendus eos quis, minima eligendi voluptatibus dolores distinctio illo impedit rerum deserunt modi dolorem, aliquam dolorum recusandae eveniet vero sint beatae repellat, non aperiam cumque. Voluptate tempora libero optio molestiae quis sapiente, molestias aliquam expedita eaque iste accusantium sint fugiat nisi culpa quibusdam quae. Laborum possumus odit eius iusto id harum nisi fugit, fugit iste exercitationem totam illo itaque a error maxime, impedit et

nihil deleniti aliquid incidunt, libero in deserunt sint nam exerci-
tationem voluptas consetetur, id voluptatum iure pariatur autem
tempora?